nBRDMs from Pauli measurements on MIMIQ

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On a quantum computer the state vector cannot be read out, and only the expectation values of Pauli strings can be measured. Here we check whether the nBRDMs, and therefore the GME concurrence, can be obtained from these measurements alone. As a first test we use the half-filled Hubbard ring, for which the exact results are already available from the thesis code.

The nBRDM extracted from the orbitals $\phi_1, \ldots, \phi_n$ is

$$\Gamma^{(n)}_{\boldsymbol{\sigma},\boldsymbol{\sigma}'} = \langle \Psi_e |\, c^\dagger_{n,\sigma'_n} \cdots c^\dagger_{1,\sigma'_1} c_{1,\sigma_1} \cdots c_{n,\sigma_n} \,| \Psi_e \rangle, \tag{1}$$

and the spin state is $\rho_s = \Gamma^{(n)}/\operatorname{Tr}\Gamma^{(n)}$. Since $\phi_1, \ldots, \phi_n$ are different orbitals, the operator in Eq. (1) can be rearranged into one pair for each orbital,

$$c^\dagger_{n,\sigma'_n} \cdots c^\dagger_{1,\sigma'_1} c_{1,\sigma_1} \cdots c_{n,\sigma_n} = \big(c^\dagger_{1,\sigma'_1} c_{1,\sigma_1}\big)\big(c^\dagger_{2,\sigma'_2} c_{2,\sigma_2}\big) \cdots \big(c^\dagger_{n,\sigma'_n} c_{n,\sigma_n}\big). \tag{2}$$

For $n = 2$, for example, $c_2^\dagger c_1^\dagger c_1 c_2 = \big(c_1^\dagger c_1\big)\big(c_2^\dagger c_2\big)$: the operator $c_2^\dagger$ is moved to the right past $c_1^\dagger$ and c_1, and the two minus signs cancel. In general the number of exchanges is always even, so no sign appears. This is the form used in the code. With the Jordan–Wigner transformation, $c_j = Z_0 \cdots Z_{j-1}(X_j + iY_j)/2$, every element then becomes a linear combination of Pauli strings,

$$\Gamma^{(n)}_{\boldsymbol{\sigma},\boldsymbol{\sigma}'} = \sum_k a_k \, \langle \Psi_e | P_k | \Psi_e \rangle, \tag{3}$$

where the coefficients a_k are known from the transformation. For the Hubbard ring the orbital ϕ_k is localized on site k, and it corresponds to the qubits $2k$ (spin up) and $2k+1$ (spin down), with the sites counted from 0. For a single orbital, for example, we have $\Gamma_{\uparrow\uparrow} = \frac{1}{2} - \frac{1}{2}\langle Z_0 \rangle$ and $\Gamma_{\uparrow\downarrow} = \frac{1}{4}(\langle X_0 X_1 \rangle + \langle Y_0 Y_1 \rangle) - \frac{i}{4}(\langle X_0 Y_1 \rangle - \langle Y_0 X_1 \rangle)$, so that six Pauli strings are needed.

In practice, three things are passed to MIMIQ. The first is the Hamiltonian H_e of the ring, built with the thesis code and mapped to Pauli strings by Jordan–Wigner; this is only used to check that $\langle H_e \rangle = E_0$. The second is the state $|\Psi_e\rangle$, which for now is the exact ground state, loaded into the Exaqt simulator as a vector of 2^{2L} amplitudes. Since MIMIQ takes qubit 0 as the lowest bit, while OpenFermion takes it as the highest, the bit order of the vector is reversed before loading. The third is the list of Pauli strings P_k: for each of them we add one expectation-value instruction to the circuit, so that all strings are measured in a single run. Strings that appear in more than one cluster are measured only once. What MIMIQ gives back is one real number for each string, $z_k = \langle \Psi_e | P_k | \Psi_e \rangle$, and nothing else. The nBRDM is then built outside MIMIQ from Eq. (3), and normalized to unit trace.

We tested this for $t = 1$, $U = 4$ and $\epsilon = V = 0$. The energy measured on MIMIQ agrees with the exact one to 2×10^{-15} for $L = 4$ ($E_0 = -2.1027484835$) and to 3×10^{-15} for $L = 6$ ($E_0 = -3.6687061789$). The nBRDMs agree with those computed directly by the thesis code to machine precision, for all clusters up to $n = 4$ (Table 1). The number of Pauli strings grows roughly as 7^n per cluster; on the exact simulator this is not a problem (0.4 s for all the strings with $n = 4$ and $L = 6$), but it will become the main cost when the expectation values are estimated from a finite number of shots.

As a side remark, the code stores the matrix with the two spin indices exchanged, i.e. $(\Gamma^{(n)})^T$. Since $\Gamma^{(n)}$ is Hermitian this is its complex conjugate, which has the same GME concurrence, and for the Hubbard ground state the two coincide because $\Gamma^{(n)}$ is real.

The next step is to compute $\mathcal{C}_{\mathrm{GME}}$ from these nBRDMs with the GCG solver and to compare it with the thesis values. After that, the exact state will be replaced by a state prepared on MIMIQ with VQE, and later the same procedure will be applied on the digital twin.

	$L = 6$		max. difference	
n	clusters	Pauli strings	$L = 4$	$L = 6$
1	6	37	4×10^{-15}	2×10^{-16}
2	15	577	4×10^{-15}	3×10^{-16}
3	20	4 897	4×10^{-15}	4×10^{-16}
4	15	24 337	4×10^{-15}	3×10^{-16}

Table 1: Number of clusters and of distinct Pauli strings, and largest difference between the nBRDMs obtained from the MIMIQ measurements and the exact ones. All the combinations of n sites are included; the consecutive clusters are a subset of these.